Phase 3: UI Design & Frontend Implementation

CMPS 4200 HCI: AI-Enhanced UI/UX | Fall 2026

Duration	Weight	Presentations
4 weeks (weeks of Oct 5 through Nov 2)	25% of final grade	Presentation, in class, week of Nov 2

Objective

Design your application interface in high fidelity, then implement it. Every team does both. Design your core screens in Figma (or a similar tool) and start building once they take shape, refining them and designing secondary screens as you go.

This phase ends with the **midterm presentation**, which also serves as a dress rehearsal for the final presentation. The user testing you do here sets a **baseline**: in Phase 5 you will repeat the same tasks and measures to show how much your design improved.

Your AI calls are not connected yet. You will build against a **mock AI service** so that every AI state can be designed, built, and tested now; in Phase 4 you replace the mock with real API calls.

Key Dates

When	What happens
Week of Oct 5	Phase 3 kickoff. Start the visual design system and high-fidelity screens in Figma.
Week of Oct 19 (in class)	**Figma design update:** a 5-10 minute per-team update showing your design.
Week of Oct 26 (in class)	**Cross-team usability testing** on your running frontend. Bring a working build; each team tests with members of other teams.
Week of Nov 2	**All deliverables due before class.** Midterm presentations in class.

Grade Breakdown

Component	Weight
1. Visual Design System and Screen Designs	30%
2. Frontend Implementation	30%
3. User Testing and Iteration	30%
4. Midterm Presentation	10%

1. Visual Design System and Screen Designs (30%)

- **Style Basics:** Before designing screens, decide and write down: your color palette (background, text, one accent, plus error and success colors), your font choices and 4–5 text sizes, and one spacing scale (for example 4, 8, 16, 24, 32). Use these same values on every screen. You will reuse them in code.
- **Screen Designs:** All major screens from your Phase 2 site map, fully designed in high fidelity.
- **AI Components:** Design each AI feature in all five states from Phase 2:
  - Empty/initial
  - Loading (matched to the latency you measured)
  - Success

  - Low-confidence or uncertain result
  - Error, with a clear path to recovery
- **User Control:** Show how users edit, reject, regenerate, or undo AI results.
- **Changes from Phase 2:** Briefly list what changed from your wireframes and why (for example, peer critique or instructor feedback).

3. Frontend Implementation (30%)

All teams implement. Use React or another modern framework.

- **At least 4-5 core screens** built from your Figma designs, with working navigation and sample data.
- **Design system in code:** colors, typography, and spacing defined once (for example, as tokens, variables, or a theme) and reused, not hard-coded per screen.
- **Mock AI service:** put all AI calls behind a single service module that currently returns mock data. It must be able to return a success, a slow response, and an error, so that all five AI states can be shown and tested. In Phase 4 you replace the mock with real API calls.
- **User control implemented:** users can edit, reject, regenerate, or undo AI results.
- **Repository:** a Git repository with meaningful commits from every team member and a README explaining how to run the app. API keys must never be committed.

> **Why a mock service?** It lets you build and test your loading, error, and low-confidence states now, without waiting on API keys, rate limits, or the live model. Teams that do this well mostly swap one file in Phase 4; teams that skip it rebuild their screens.

4. User Testing and Iteration (30%)

Running the Tests

- Conduct **5-6 user tests** on your running frontend. In-class cross-team tests count.
- Use a written test script with **at least 3 tasks** taken from your Phase 2 primary flows, and use the same script for every participant.
- Ask participants to think aloud. At least one task should lead the participant into an AI error or low-confidence result (use your mock service to trigger it).
- **Consent:** read a short consent script before each session, record audio or video only with permission, and do not collect personal or sensitive information.

What to Measure

- **Task success rate** and **time on task** for each task.
- **System Usability Scale (SUS)** questionnaire after each session.
- **AI trust:** at least one question on how much participants trust the AI results (for example, a 1-5 rating plus a short explanation).
- Keep your script and results: Phase 5 repeats these tasks and measures for a before-and-after comparison.

Test Report and Iteration

- Summarize findings and rate each issue by severity (0 = not a problem, 4 = usability catastrophe).
- Implement fixes for the most severe issues before the midterm presentation, and show before-and-after screens.
- List the remaining issues you plan to address in Phase 4.

> **Tip:** Your SUS scores and task success rates are your baseline. Record them carefully; being able to show measurable improvement in Phase 5 is one of the strongest parts of a final presentation.

6. Midterm Presentation (10%)

Each team has **10 minutes plus 2 minutes of Q&A**, strictly timed. Every team member should speak. Because the demo is live, have a short recorded backup ready in case something fails.

AI Tool Use

You are encouraged to use AI tools (for example, Claude, ChatGPT, Copilot, Figma AI, v0) for design, implementation, and coding. The following conditions apply:

- You remain responsible for all work you submit. Any team member may be asked to explain any part of your design or code.
- **User research and testing must involve real participants.** AI-generated or simulated users, test sessions, or data are not permitted.
- Do not paste API keys, or participants' personal information or recordings, into AI tools.
- Include a brief statement in your submission describing which AI tools you used and for what.

What to Submit

- A single PDF containing, screen designs, test script, test report, list of changes, and AI tool use statement.
- A link to your Figma (or similar) file with view access for the instructor.
- Repository link, with the README and mock AI service. No API keys included.
- Raw test data (task results, times, and SUS responses) as a spreadsheet.
- Presentation slides.
- A team contribution statement: a short list of what each member contributed, agreed to by all members. This may be used to adjust individual grades.